# Experiment 3 : Data Cleaning and Preparation

**Aim:** To perform data cleaning and preparation on a telecom customer churn dataset to gain insi
the factors that contribute to customer churn.

In [1]:
```
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import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
```

In [2]:
```
np.random.seed(2)
n = 300
df = pd.DataFrame({
    "customer_id": range(1, n + 1),
    "gender": np.random.choice(["Male", "Female", "male ", "FEMALE"], n),
    "age": np.random.randint(18, 70, n),
    "tenure": np.random.randint(1, 72, n),
    "monthly_charges": np.round(np.random.uniform(20, 120, n), 2),
    "total_charges": np.round(np.random.uniform(100, 8000, n), 2),
    "contract": np.random.choice(["Month-to-month", "One year", "Two year
    "churn": np.random.choice(["Yes", "No"], n),
})
df.head()
```

Out[2]:

| | customer_id | gender | age | tenure | monthly_charges | total_charges | contract |
|---|---|---|---|---|---|---|---|
| 0 | 1 | Male | 51 | 21 | 117.92 | 6825.09 | Two year |
| 1 | 2 | FEMALE | 56 | 28 | 28.16 | 6095.94 | One year |
| 2 | 3 | Female | 25 | 32 | 55.84 | 3074.01 | Two year |
| 3 | 4 | Male | 50 | 10 | 108.77 | 4773.68 | Month-to-month |
| 4 | 5 | male | 67 | 21 | 82.40 | 2290.87 | Two year |

In [3]:
```
df.loc[np.random.choice(n, 15), "age"] = np.nan
df.loc[np.random.choice(n, 10), "monthly_charges"] = np.nan
df.loc[np.random.choice(n, 5), "monthly_charges"] = 9999
df = pd.concat([df, df.iloc[:8]], ignore_index=True)
print("shape after adding missing values, outliers and duplicates:", df.s
```

```
shape after adding missing values, outliers and duplicates: (308, 8)
```

In [4]:
```
print("missing values:")
print(df.isnull().sum())
print("duplicate rows:", df.duplicated().sum())
```

```
missing values:
customer_id         0
gender              0
age                15
tenure              0
monthly_charges     8
total_charges       0
contract            0
churn               0
dtype: int64
duplicate rows: 8
```

In [5]:
```
df = df.drop_duplicates()
print("rows after removing duplicates:", len(df))
```

```
rows after removing duplicates: 300
```

In [6]:
```
df["age"] = df["age"].fillna(df["age"].median())
print("missing age now:", df["age"].isnull().sum())
```

```
missing age now: 0
```

In [7]:
```
df["monthly_charges"] = df["monthly_charges"].fillna(df["monthly_charges"]
print("missing monthly_charges now:", df["monthly_charges"].isnull().sum(
```

```
missing monthly_charges now: 0
```

In [8]:
```
df["gender"] = df["gender"].str.strip().str.lower()
df["age"] = df["age"].astype(int)
print("gender values now:", list(df["gender"].unique()))
```

```
gender values now: ['male', 'female']
```

In [9]:
```
q1 = df["monthly_charges"].quantile(0.25)
q3 = df["monthly_charges"].quantile(0.75)
iqr = q3 - q1
df = df[(df["monthly_charges"] >= q1 - 1.5 * iqr) & (df["monthly_charges"
print("rows after outlier removal:", len(df))
```

```
rows after outlier removal: 295
```

In [10]:
```
df["avg_monthly"] = df["total_charges"] / df["tenure"]
df["is_senior"] = (df["age"] >= 60).astype(int)
df.head()
```

Out[10]:

| | customer_id | gender | age | tenure | monthly_charges | total_charges | contract | chu |
|---|---|---|---|---|---|---|---|---|
| 0 | 1 | male | 51 | 21 | 117.92 | 6825.09 | Two year | |
| 1 | 2 | female | 56 | 28 | 28.16 | 6095.94 | One year | Y |
| 2 | 3 | female | 25 | 32 | 55.84 | 3074.01 | Two year | |
| 3 | 4 | male | 50 | 10 | 108.77 | 4773.68 | Month-to-month | Y |
| 4 | 5 | male | 67 | 21 | 82.40 | 2290.87 | Two year | |

In [11]:
```
num = ["age", "tenure", "monthly_charges", "total_charges", "avg_monthly"
df[num] = StandardScaler().fit_transform(df[num])
df.head()
```

Out[11]:

| | customer_id | gender | age | tenure | monthly_charges | |
|---|---|---|---|---|---|---|
| 0 | 1 | male | 0.396141 | -0.645453 | 1.785469 | 1 |
| 1 | 2 | female | 0.742296 | -0.301134 | -1.446509 | 0 |
| 2 | 3 | female | -1.403861 | -0.104380 | -0.449838 | -0 |
| 3 | 4 | male | 0.326911 | -1.186527 | 1.456006 | 0 |
| 4 | 5 | male | 1.503835 | -0.645453 | 0.506505 | -0 |

In [12]:
```
df = pd.get_dummies(df, columns=["gender", "contract"], drop_first=True)
print("shape after encoding:", df.shape)
```

```
shape after encoding: (295, 11)
```

In [13]:
```
train, test = train_test_split(df, test_size=0.2, random_state=2)
print("train shape:", train.shape, " test shape:", test.shape)
```

```
train shape: (236, 11)  test shape: (59, 11)
```

In [14]:
```
train.to_csv("churn_cleaned.csv", index=False)
print("done")
```

```
done
```